minimum depth

7

no proper depth-6 schedule → depth 7 is optimal

6

3	2	2	1	1	0	**9**

parity — $F_L + F_R = 9$ is odd;
$F_L = F_R$ needs even



rigidity — one X and one Z
in each of $\{1, 2\}$, $\{3, 4\}$, $\{5, 6\}$

1	2	3	4	5	6

support symmetry — X-edges lie in $[a, b]$,
$a + b = 7$
(3,4) empty · (2,5) $F_L \neq F_R$ · (1,6) continues ↑

depth 6 attainable — the exact minimum

c=0	3	6	1	4	2	5	5	3	2	1	4	6
c=1	5	4	2	1	3	6	4	6	1	3	2	5

template on $\mathbb{Z}_2$, lifted along ϵ
(ord(d) even, 2-height ≤ 8)



one repeated sum — four
corners cancel: $1 + 0 + 0 + 1 \equiv 0$

$M_\pi = 0 \Leftrightarrow$ proper: every overlapping pair has an even number of X-first corners

distinct sums $\Rightarrow F_L = F_R$

same check weight, same qubit degree — only the algebra of A and B changed